

# ELEANOR LEE

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## PROFESSIONAL HIGHLIGHTS

Senior Data Scientist (9+ years) building measurement and decision systems for always-on marketing, and productionizing causal methods into reusable AI-native infrastructure. Productionized a propensity-score-matching analytical skill on the Claude Agent SDK with human-in-the-loop guardrails, adopted across Facebook, Threads, and WhatsApp; architected Instagram's first Connected TV launch (\$40M, matched-market geo holdout capturing ~80% of incremental installs); and built a brand-preference uplift framework (~320K respondents) informing 11 product experiments and 3 targeting strategies. Earlier shipped 3 GenAI products end-to-end at Allianz (OpenAI API, RAG, LLM evaluation). Specialize in leading-indicator design, holdouts for non-randomizable always-on environments, and decision loops that turn analysis into shipped changes — comfortable embedded with PMs and engineers in 0→1, B2C and SMB-scale settings.

## SKILLS AND TECHNIQUES

- **Languages:** SQL, Hive, Python (Pandas, NumPy, SciPy, Scikit-Learn, PyTorch, statsmodels, lifelines), R, PySpark, Bash
- **Causal Inference & Measurement:** MMT, DiD, ITS, Synthetic Control, PSM; Uplift / CATE (X-Learner, CausalML); MMM; MTA; channel incrementality (matched-market geo & nested holdouts, temporal switch-off); CUPED; sequential inference; Brand Lift Studies; survey weighting (post-stratification, MrP, raking); clean room, MMP (AppsFlyer)
- **Experimentation & Decision Systems:** Multi-cell RCT, nested holdout, Bonferroni, multi-armed bandit (Thompson, UCB), power / MDE, pre-committed decision loops, next-best-treatment prioritization, leading-indicator frameworks
- **Agentic & AI-Native:** OpenAI API, Anthropic Claude (API, Agent SDK, MCP), LangChain, LlamaIndex, FAISS, Whisper, RAG, LLM evaluation (MuSR, GLUE, similarity), HITL guardrails; AWS (SageMaker, Bedrock), GCP (BigQuery, Vertex AI), Azure (OpenAI, AI Search, Prompt Flow), Databricks

## WORK EXPERIENCE

**Meta — Senior Data Scientist, Instagram Consumer Marketing Data Science**  
*Senior Data Science & Marketing Manager, Instagram Product GTM and Brand*

San Francisco, CA  
Aug 2025 – Current

### AI-Native Workflow Automation & Cross-Team Methodology Adoption

*Productionize measurement methodologies into reusable AI-powered infrastructure adopted across the Marketing DS organization.*

- **PSM AI Skill (Claude Agent SDK):** Build and ship a multi-step analytical skill wrapping propensity score matching — automated data prep, propensity scoring, and quality gates (SMD balance, A/A validation, overlap diagnostics) with HITL guardrails; cut turnaround from 2–4 weeks to days, adopted by Facebook, Threads, and WhatsApp
- **Real-Time In-Flight Tracker:** Build an automated Hive-connected tracker surfacing CTR, VTR, engagement, CPI, and CPM; shift budget optimization from a 2-week post-campaign lag to a 7-day in-flight cadence — the leading-indicator framework enabling weekly iteration.

### Instagram Brand Preference — Upper-Funnel Causal Measurement & Uplift Targeting

*Own causal measurement and audience targeting for the brand-preference north-star metric. \$17M budget, ~1B impressions, 5 multi-cell experiments.*

- **Causal ML & Uplift Modeling:** Build a propensity + uplift framework (LightGBM + X-Learner CATE) on ~320K respondents; identify the persuadable middle (+2.8–4.1pp) and suppress high-baseline segments (–7.4pp); translate into 3 deployment tiers informing 11 experiments and 3 targeting strategies, reviewed by Directors from DS and Research.
- **Pre-Committed Decision Loop:** Author a 5-scenario outcome-to-action decision matrix (clear winner, hidden subgroup, Simpson's paradox, true null) that pre-commits leadership to downstream strategy before results land.

### Instagram Connected TV (CTV) Launch — Always-On Measurement Architecture (0→1)

*Designed the end-to-end measurement system for Instagram's first-ever CTV launch — \$40M media budget*

- **Channel Incrementality:** Establish matched-market geo-holdout incrementality tests capturing ~80% of incremental paid-ad installs where user-level randomization wasn't feasible; drive weighted CPI from \$10.50 to \$3.30 and a 7% adoption lift via placement-mix re-ranking and frequency-cap optimization.
- **Hybrid Measurement Stack:** Architect a 3-source attribution stack — clean rooms, MMP, and first-party deep-link tracking — trading speed against accuracy, with time to conversion analysis and path to conversion analysis to define optimal MTA lookback-window (1/7/14-day click, 1/5/12-hour view-through).

**Apple**  
*Lead Data Science Consultant (Contract from Infosys) — App Store, Marketing Analytics*

Los Angeles, CA  
Dec 2024 – Aug 2025

Led measurement for \$22M+ in paid App Store campaigns globally; built an iROAS / ROAS simulation across \$1M–\$7M tiers and set the team standard for Matched Market Tests, propensity-matched controls, and Synthetic Control.

- **End-to-End ML Measurement Pipeline:** Productionized 3 pipelines — attribution, Matched Market Tests (GeoLift + CausalImpact), and Synthetic Control matching — with dynamic calibration and fail-safe logging; cut MMT turnaround from 5 weeks to 8 days, reduced manual QA 40%, reduced SCM posterior variance ~35%, adopted across 12+ global campaigns.
- **Creative × Audience Experiments:** Designed 3-cell experiments separating attributed ROAS from incremental iROAS by creative and audience; surfaced distinct lift profiles and redirected ~\$2.5M in seasonal spend toward higher-iROAS pairings.
- **Cold-Start Estimator (0→1):** Built a PSM + Synthetic Control hybrid estimator for new apps with no history, matching across 10K+ apps to construct counterfactual peer sets; enabled go/no-go GTM investment and supported \$30M+ growth planning.

Allianz Life

Remote

Senior Data Scientist, GenAI and GTM, Center of Excellence

Sep 2023 – May 2024

- **RAG Chatbot + Causal Impact:** Shipped an LLM-powered RAG chatbot for annuity agents (LangChain, OpenAI API, FAISS, LlamaIndex); quantified a 30% productivity lift and \$15–20M annual incremental revenue via Difference-in-Differences, driving firm-wide rollout.
- **AI Products + Voice-to-Text Prototype:** Built Document GPT (presented to 60+ executives) and a 0→1 Enterprise Knowledge Assistant (3-week POC, Azure AI Search + Prompt Flow); prototyped a voice-to-text legal assistant (Whisper + OpenAI API + LlamaIndex, chain-of-thought).
- **LLM Evaluation Framework:** Designed the firm's first LLM evaluation harness (MuSR, GLUE, semantic similarity, precision/recall) anchored to compliance policies; drove a 15% performance improvement and 20% critical-error reduction.

Citigroup

New York City, NY

Data Scientist, Test and Learn → Lead Data Analyst, Omnichannel Customer Journey Analytics

Nov 2021 – Jun 2023

- **Multi-Armed Bandit Testing:** Launched an optimized-CTA pilot for the iOS dispute tracker via multi-armed bandit testing; drove a 30% increase in feature adoption within 3 months.
- **Real-Time Journey Platform & Driver Analysis:** Partnered with Data Engineering on a real-time omnichannel analytics platform sourcing 70+ systems via PySpark + Hive with CI/CD (3K+ metrics); built a 22K-feature XGBoost attrition model driving a 431% lift in retained balances by targeting the top 13% of at-risk customers.

The Home Depot, Services

Atlanta, GA

Senior Data Analyst — Data Science, Services Sales

Aug 2020 – Nov 2021

- **Predictive Lead Scoring & MTA:** Built a lead-scoring model (LightGBM, Logistic Regression) integrated with Salesforce CRM workflows; increased connect rate 17% via A/B testing and designed a multi-touch attribution model for lifecycle marketing.

Hawyang Semiconductor Corporation

Taipei, Taiwan

Product & Sales Analytics Manager — Connected Car

Oct 2017 – Jun 2019

- Led weekly competitor and sales intelligence reporting (Tableau + SQL) and built time-series forecasting with outlier detection for 30K+ SKUs; insights drove 17% QoQ growth and supported dynamic-pricing decisions.

EDUCATION

Emory University, Goizueta Business School

Atlanta, GA

Master of Science in Business Analytics (STEM)

May 2020

National Taiwan University (NTU)

Taipei, Taiwan

Bachelor of Business Administration, Finance

Jun 2017